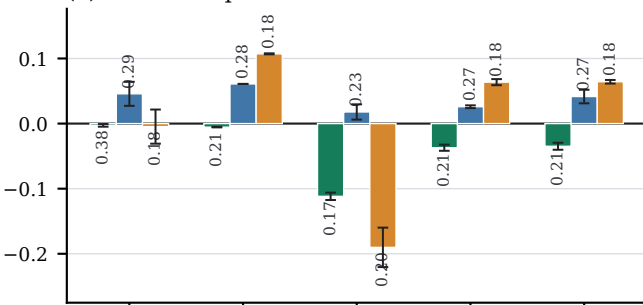


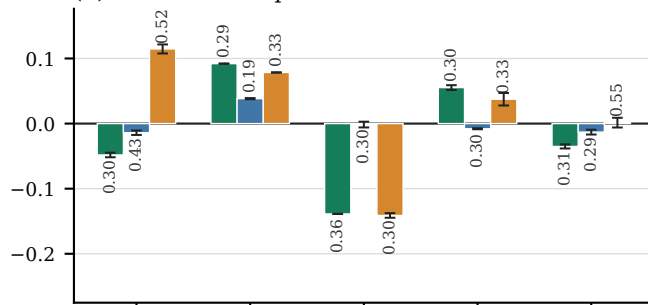
ARIMA + LSTM    ARIMA + XGBoost    XGBoost + LSTM

Sharpe difference vs. stronger

(a) SSE50: Representative models

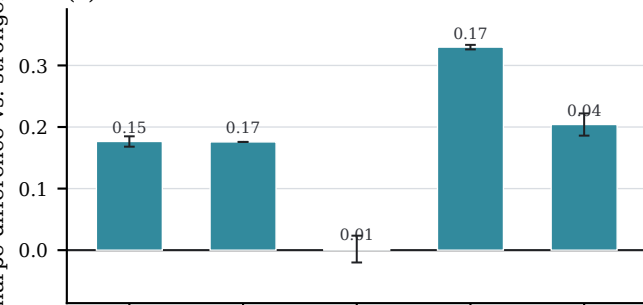


(b) HSTech10: Representative models

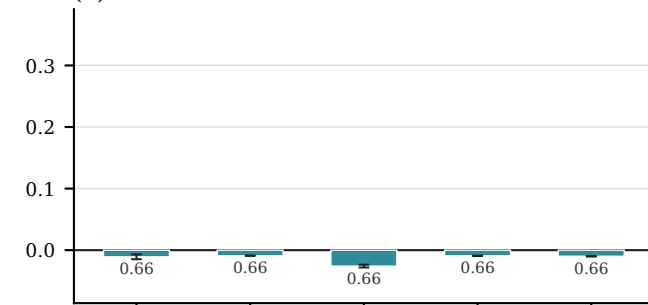


Sharpe difference vs. stronger

(c) SSE50: PatchTST+iTransformer

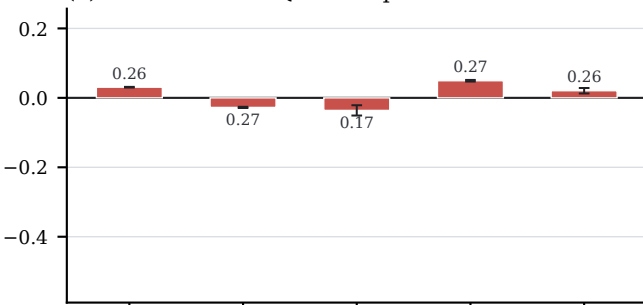


(d) HSTech10: PatchTST+iTransformer

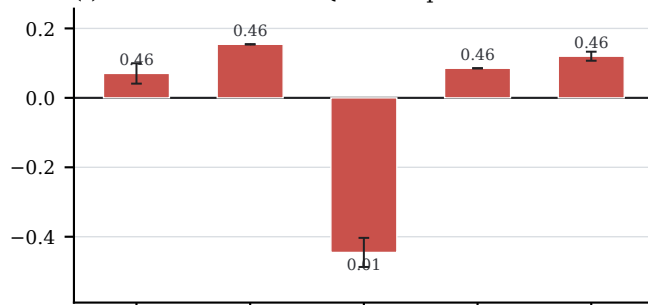


Sharpe difference vs. stronger

(e) SSE50: PPO+TQC main pair

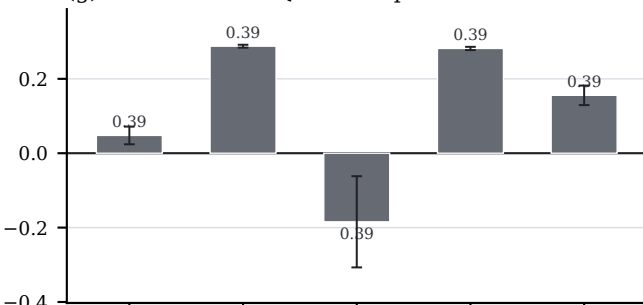


(f) HSTech10: PPO+TQC main pair

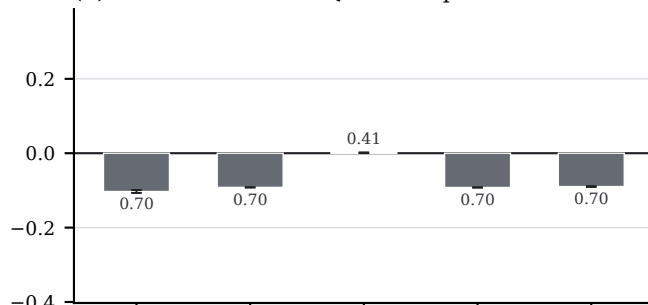


Sharpe difference vs. stronger

(g) SSE50: TD3+TQC stress pair



(h) HSTech10: TD3+TQC stress pair



Classifier group

Classifier group

Difference is relative to the stronger component; bar labels are selected fixed global thresholds.